

North Carolina State University
CH 223, Spring 2023
Organic Chemistry II

Mon-Wed-Fri, 11:45am-12:35pm, in-person lecture at 222 Dabney Hall

Course Description: Second half of two-semester sequence in the fundamentals of modern organic chemistry. Structure and bonding, stereochemistry, reactivity and synthesis of organic compounds. Detailed coverage of aromatic hydrocarbons, conjugated systems and condensation reactions. (Tentative course schedule is available in page 8.)

Learning Outcomes: By the end of the course, students will be able to:

1. Draw structures and name organic compounds including benzene derivatives, carbonyl compounds and nitrogen containing compounds.
2. Describe the attributes that make a molecule aromatic.
3. Describe the bonding, polarity, and the boiling point and solubility trends in the compounds listed in #1.
4. Compare the relative stability and reactivity of various carbonyl compounds.
5. Draw the mechanisms, including transition states and intermediates, and predict the products of electrophilic and nucleophilic aromatic substitution reactions, carbonyl addition/substitution and reduction reactions, carbonyl alpha-C reactions, alcohol oxidation reactions, condensation reactions and saponification.
6. Correlate the results of various organic reactions with predictions from molecular orbital theory.
7. Show the synthetic use of various reagents in functional group transformations.
8. Devise synthetic schemes for preparing various organic compounds.
9. Predict the products for the reactions of various functional groups.
10. Describe the functional groups and reactions of major importance in biochemistry

Instructor: Jun Ohata, Ph.D. (johata@ncsu.edu), 527A Dabney

Teaching assistant: This course does not have a teaching assistant. If you have questions, please talk to Dr. Ohata. If your questions are not personal, please consider using Student Help Forum in the Moodle page to post your question, so that other students with the same question can also know the answer.

Prerequisites: CH 221 (with a grade of B or better) or CH 225 and CH 222 or CH226.

Corequisite: CH 228

Laboratory: CH 224 is the laboratory course which should be taken concurrently with CH223 and **is administered separately**. Refer all inquiries about CH228 Laboratory to Dr. Maria Gallardo-Williams (mtgallar@ncsu.edu) 252 Fox Hall, 515-2539.

Course Materials:

1. Textbook: Carey, Giuliano, *Organic Chemistry*, 11th ed. Edition (\$40-\$160). Older/newer editions can be used (make sure to have matching sets of text and solutions manual). Students are welcome to use additional organic chemistry textbooks and online materials. (Lecture slides are based on Loudon, Parise organic chemistry which is very similar to Carey/Giuliano textbook.)
2. **Achieve (~40\$, required)** See the last page of the syllabus for the instructions. <https://achieve.macmillanlearning.com/courses/arr83h>
3. **Chemistry Coach** (<https://chemistry.coach/>). Free online chemistry website.
4. Molecular model kits (optional).



Attendance: Students are expected to attend all in-person lectures. Missing lecture(s) will put any students at a serious disadvantage with respect to learning course material. It is unnecessary to report the absence for the lectures to the instructor, but if a student wishes to keep the record of the absence, please submit Google Form instead of email the instructor (https://docs.google.com/forms/d/e/1FAIpQLScn9NiD0n8pBa4UK4hf72QaDOtNL-vz4EN2YHcCYLBwqvXZlw/viewform?usp=sf_link).

Course Grading:

Exam I: 200 pts., 20% (02/08/2023, 11:45am-12:35pm)
Exam II: 200 pts., 20% (03/10/2023, 11:45am-12:35pm)
Exam III: 200 pts., 20% (04/12/2023, 11:45am-12:35pm)
Final exam: 250 pts., 25% (5/1/2023, 12:00pm-2:30pm, Dabney 222)
Homework assignments (Achieve): 150 pts., 15% (8 assignments, about 2% each)

This course uses standard NC State letter grading:

Points	Grade	Points	Grade
970 +	A+	730 – 769	C
930 – 969	A	700 – 729	C-
900 – 929	A-	670 – 699	D+
870 – 899	B+	630 – 669	D
830 – 869	B	600 – 629	D-
800 – 829	B-	0 – 599	F
770 – 799	C+		

Extra credit/Curves: If extra credit is given for any assignment, the maximum score possible for that assignment will be a 100. Any “excess” extra credit points will not be applied to other assignments. There will be no curve applied to any homework assignment.

Re-Grade Policy: We are happy to re-grade any question you feel has been improperly graded. However, before turning in your test for a re-grade, please be aware of the following:

- 1) You have seven days from the time the test is returned to request a re-grade. After that time frame has expired, we will no longer entertain re-grade requests.
- 2) To protect the integrity of the class, tests are randomly photocopied and placed on file prior to being returned. If you return an exam for a re-grade in which the answer to the question you want re-graded has been changed in an effort to get additional points, you will be immediately turned over to the office of student conduct for disciplinary action.
- 3) Only exams written in pen can be regraded. Pen copied over in pencil will not be accepted.

Missed Exam and Homework Policy:

Excuses for *unanticipated* absences must be reported to Dr. Ohata as soon as possible. If you are incapable of taking an exam, a university-approved absence can be used for ONE exam (e.g. doctor’s note). You must fill out and submit the absence verification form at the bottom of this page: <https://dasa.ncsu.edu/support-and-advocacy/find-help/absence-verification/>. If your absence is verified and approved, you will be given a make-up exam exactly one week after the missed in-class exam during class time. If you miss an exam and do not have a University-

approved excuse, you will be given zero for that exam. Excuses for *anticipated absences* must be cleared with Dr. Ohata before the absence. Exams can be taken at an alternate date if 2-week advance notice is provided. Missing more than one in-class exam will result in an incomplete (IN) for this class.

University-excused absence information: <https://policies.ncsu.edu/regulation/reg-02-20-03-attendance-regulations/>

Course Status:

Audits: To obtain a grade of AU in the course, the student must take ALL exams. Information about and requirements for auditing a course can be found at: <http://policies.ncsu.edu/regulation/reg-02-20-04>.

Credit Only: In order to receive a grade of S, students are required to take all exams and quizzes, complete all assignments, and earn a grade of C- or better. Conversion from letter grading to credit only (S/U) grading is subject to university deadlines. Refer to the Registration and Records calendar for deadlines related to grading. For more details refer to: <http://policies.ncsu.edu/regulation/reg-02-20-15>.

Changing your Status (or dropping):

January	13	Friday	Last day to add a course without permission
January	23	Monday	Census Date/Official Enrollment Date
March	6	Monday	Drop/Revision Deadline

A full description of the College of Sciences course policy can be found at the following websites: <https://sciences.ncsu.edu/academics/academic-policies-and-procedures/> and <https://studentservices.ncsu.edu/your-classes/withdrawal/process/>. Dropping specific courses (or changing to CR or AU) after this deadline is considered only for unforeseen and unavoidable extenuating personal situations (***not poor performance***).

Incomplete Grades: If an extended deadline is not authorized by the instructor or department, an unfinished incomplete grade will automatically change to an F after either (a) the end of the next regular semester in which the student is enrolled (not including summer sessions), or (b) the end of 12 months if the student is not enrolled, whichever is shorter. Incompletes that change to F will count as an attempted course on transcripts. The burden of fulfilling an incomplete grade is the responsibility of the student. The university policy on incomplete grades is located at <http://policies.ncsu.edu/regulation/reg-02-50-03>.

Academic Integrity: Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct found at <http://policies.ncsu.edu/policy/pol-11-35-01>. Cheating or plagiarism will result in failure of this course.

Students with Disabilities: Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, student must register with the Disability Resource Office (<https://dro.dasa.ncsu.edu>), 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation at: <http://policies.ncsu.edu/regulation/reg-02-20-01>.

Non-Discrimination Statement: NC State University provides equality of opportunity in education and employment for all students and employees. Accordingly, NC State affirms its commitment to maintain a work environment for all employees and an academic environment for

all students that is free from all forms of discrimination. Discrimination based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or NC State University policy and will not be tolerated. Harassment of any person (either in the form of quid pro quo or creation of a hostile environment) based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation also is a violation of state and federal law and/or NC State University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at <http://policies.ncsu.edu/policy/pol-04-25-05> and <https://oied.ncsu.edu/divweb>. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.

Supporting Fellow Students in Distress: As members of the NC State Wolfpack community, we each share a personal responsibility to express concern for one another and to ensure that the classroom and the campus as a whole remains a safe environment for learning. Occasionally, you may come across a fellow classmate whose personal behavior concerns or worries you. If this is the case, you are encouraged to report this behavior to the NC State Cares website: <https://ncstatecares.dasa.ncsu.edu/>. Although you can report anonymously, it is preferred that you share your contact information so they can follow-up with you personally.

Course Evaluations: Online class evaluations will be available for students to complete during the last two weeks of class. Students will receive an email message directing them to a website where they can login using their Unity ID and complete evaluations. All evaluations are confidential; instructors will never know how any one student responded to any question, and students will never know the ratings for any particular instructors. Evaluation website: <https://oirp.ncsu.edu/classeval/>

COVID-related policy

Due to the COVID-19 pandemic, public health measures continue to be implemented across campus. Students should stay current with these practices and expectations through the [Protect the Pack](https://www.ncsu.edu/coronavirus/) website (<https://www.ncsu.edu/coronavirus/>). The sections below provide expectations and conduct related to COVID-19 issues.

Health and Participation in Class

We are most concerned about your health and the health of your classmates and instructors/TAs.

- If you test positive for COVID-19, or are told by a healthcare provider that you are presumed positive for the virus, you should not attend any face-to-face (F2F) classes or face-to-face component of a hybrid class. Work with your instructor on any adjustments necessary; also follow other university guidelines, including self reporting ([Coronavirus Self Reporting](#)): Self-reporting is not only to help provide support to you, but also to assist in contact tracing for containing the spread of the virus.
- If you feel unwell, even if you have not been knowingly exposed to COVID-19, please do not come to a F2F class or activity.
- If you are in quarantine, have been notified that you may have been exposed to COVID-19, or have a personal or family situation related to COVID-19 that prevents you from attending this course in person (or synchronously), please connect with your instructor to make alternative plans, as necessary.
- If you need to make a request for an academic consideration related to COVID-19, such as a discussion about possible options for remote learning, please talk with your instructor.

Health and Well-Being Resources

These are difficult times, and academic and personal stress are natural results. Everyone is encouraged to [take care of themselves](#) and their peers. If you need additional support, there are many resources on campus to help you:

- Counseling Center ([NCSU Counseling Center](#))
- Student Health Services ([Health Services | Student](#))
- If the personal behavior of a classmate concerns or worries you, either for the classmate's well-being or yours, we encourage you to report this behavior to the NC State CARES team: ([Share a Concern](#)).
- If you or someone you know are experiencing food, housing or financial insecurity, please see the Pack Essentials Program ([Pack Essentials](#)).

Community Standards related to COVID-19

We are all responsible for protecting ourselves and our community. Please see the [community standards](#) and Rule 04.21.01 regarding Personal Safety Requirements Related to COVID-19 [RUL 04.21.01 – Personal Safety Requirements Related to COVID-19 – Policies, Regulations & Rules](#)

Course Expectations Related to COVID-19:

- **Face Coverings:** All members of the NC State academic community are expected to follow all university policies and guidelines, including the [Personal Safety Rule](#) and [community standards](#), for the use of face coverings.

- **Course Attendance:** NC State attendance policies can be found at: [REG 02.20.03 – Attendance Regulations – Policies, Regulations & Rules](#). Please refer to the course's attendance, absence, and deadline policies for additional details. If you are quarantined or otherwise need to miss class because you have been advised that you may have been exposed to COVID-19, you should not be penalized regarding attendance or class participation. However, you will be expected to develop a plan to keep up with your coursework during any such absences. If you become ill with COVID-19, you should follow the steps outlined in the health and participation section above.
- **Technology Requirements:** This course may require particular technologies to complete coursework. Be sure to review the syllabus for these expectations, and see the [syllabus technical requirements](#) for your course. If you need access to additional technological support, please contact the Libraries' Technology Lending Service: ([Technology Lending](#)).

Course Delivery Changes Related to COVID-19

Please be aware that the situation regarding COVID-19 is frequently changing, and the delivery mode of this course could change accordingly, including from in-person to remote. Regardless of the delivery method, we will strive to provide a high-quality learning experience.

NO LONGER AVAILABLE - Grading/Scheduling Changing Options Related to COVID-19

Two policies, enhanced S/U Grading Option and Late Drop, put in place at the beginning of the COVID-19 pandemic have been discontinued.

In some cases, an option may be to request an "incomplete" in the course. If you are experiencing difficult or extenuating circumstances, you should discuss possible options with your instructor and your academic advisor.

Need Help?

If you find yourself in a place where you need help, academically or otherwise, please review these [Step-by-Step Help Topics](#). (Insert information or links for college or departmental level support programs, if available.)

Other Important Resources

- **Keep Learning:** [Keep Learning](#)
- **Protect the Pack FAQs:** [Frequently Asked Questions | Protect the Pack](#)
- **NC State Protect the Pack Resources for Students:** [Resources for Students | Protect the Pack](#)
- **Academic Success Center** (tutoring, drop in advising, career and wellness advising): [Academic Success Center](#).
- **NC State Keep Learning, tips for students opting to take courses remotely:** [Keep Learning Tips for Remote Learning](#)
- **Introduction to Zoom for students:** <https://youtu.be/5LbPzzPbYEW>
- **Learning with Moodle, a student's guide to using Moodle:** <https://moodle-projects.wolfware.ncsu.edu/course/view.php?id=226>
- **NC State Libraries** [Technology Lending Program](#)

Some other thoughts to consider for your syllabus or course website:

- Note the [syllabus regulation](#) has information about digitally-hosted components, which includes the use of zoom and also classroom recordings. Please also refer to the 'student privacy' section of the Keep Teaching FAQ: <https://provost.ncsu.edu/faculty-resources/keep-teaching/faqs/>.
- For remote exams: be very clear about what is allowed and not allowed. See, for example, the ST 311 situation where many students used Chegg. http://www.technicianonline.com/news/article_308a42f6-caf0-11ea-bc0e-2332ae88e82a.html
- If you plan to use exam proctoring: "This course may require proctored/ monitored exams. A request for in-person exam proctoring on campus has been submitted to [DELTA Testing Services](#). This information will be updated with further instructions once the availability of in-person exam proctoring on campus has been confirmed. This update may take place after the start of the semester."
- If gloves are recommended for use (such as in a computer lab): how to properly remove your gloves: https://www.youtube.com/watch?v=xTYioOo_6U
- Hand hygiene: Wash your hands frequently with soap and water for at least 20 seconds.
 - Follow proper hand-washing techniques, here is one example: <https://www.youtube.com/watch?v=bQCP7waTRWU>
 - If soap and water are not immediately available, use an alcohol-based hand-sanitizer containing $\geq 60\%$ alcohol.

Course Schedule (Subject to Change):

lecture	date			topic	chapter	Loudon
1	Jan	9	Mon	introduction	-	-
2		11	Wed	Dienes, Resonance, and Aromaticity	11	15
3		13	Fri			
		16	Mon	no class, MLK	-	-
4		18	Wed		11	15
5		20	Fri	The Chemistry of Benzene and Its Derivatives	12	16
6		23	Mon			
7		25	Wed			
8		27	Fri			
9		30	Mon			
10	Feb	1	Wed	Allylic and Benzylic Reactivity	12	17
11		3	Fri			
12		6	Mon			
		8	Wed	exam-1	11-12	15-17
13		10	Fri	Aryl Halides, Vinylic Halides, and Phenols. Metal Catalysis	13,15	18
14		13	Mon			
15		15	Wed			
16		17	Fri			
17		20	Mon	Aldehydes and Ketones. Carbonyl-Addition Reactions	18	19
18		22	Wed			
19		24	Fri			
20		27	Mon			
21	Mar	1	Wed			
22		3	Mon	The Chemistry of Carboxylic Acids	19a	20a
23		8	Wed			
24		8	Wed			
		10	Fri	exam-2	13,15,18,19a	18-20a
		13	Mon	no class, Spring break		
		15	Wed	no class, Spring break		
		17	Fri	no class, Spring break		
25		20	Mon		19b	20b
26		22	Wed	The Chemistry of Carboxylic Acid Derivatives	20	21
27		24	Fri			
28		27	Mon			
29		29	Wed			
30		31	Fri	Enolate Ions, Enols, and α,β -Unsaturated Carbonyl	21	22
31	Apr	3	Mon			
32		5	Wed			
33		7	Fri			
34		10	Mon			
		12	Wed	exam-3	19b-22	20-22
35		14	Fri	The Chemistry of Amines	22	23
36		17	Mon			
37		19	Wed			
38		21	Fri			
39		24	Mon	Course recap	-	-
	May	1	Mon	Final exam (12:00pm-2:30pm, Dabney 222)		

Note: "chapter" is for the Carey/Giuliano textbook. "Loudon" is for chapters for the Loudon/Parise textbook which the course slides are based on.

Achieve eLearning for homework and other assignments: Achieve chemistry questions (provided by MacMillan Learning) are delivered in a web browser to provide real-time grading, response-specific coaching, improvement of problem-solving skills, and detailed answer explanations. Dynamic answer modules enable one to interact with 3D models and figures, utilize drag-and-drop synthetic routes, and draw chemical structures – including stereochemistry and curved arrows.

Enrollment/registration/purchase information (**Use your NC State email address**)

1. Check if you received the invitation email from MacMillan Learning about the course. If you didn't get the invitation email, start from this link (the course ID is arr83h):
<https://achieve.macmillanlearning.com/courses/arr83h>
2. After Step 1, follow the instruction of the link below for the purchase option. It should be ~\$40.
<https://macmillan.force.com/macmillanlearning/s/article/Achieve-Join-a-course>

Once you have registered and enrolled, you can log in at any time to complete or review your homework assignments. During sign up – and throughout the term – if you have any technical problems or grading issues, use the chat system with the customer support (<https://macmillan.force.com/macmillanlearning/s/chat-with-us>). **The Achieve support team is almost always more able (and faster) to resolve issues than your instructor.**

Also please consider using the help link (<https://macmillan.force.com/macmillanlearning/s/>)

To optimize your Achieve experience, please keep your internet browser up to date and minimize the use of RAM-intensive programs/websites while using Achieve.